

```

> degree(g1, mode='all')
Amy Ram Li Kate
3 2 4 1
> diameter(g1, directed=F, weights = NA)
[1] 2
> edge_density(g1, loops = F)
[1] 0.4166667
> reciprocity(g1)
[1] 0.4
> closeness(g1, mode='all', weights = NA)
Amy Ram Li Kate
0.2500000 0.2500000 0.3333333 0.2000000
> betweenness(g1, directed=T, weights=NA)
Amy Ram Li Kate
2 0 3 0
> edge_betweenness(g1, directed=T, weights=NA)
[1] 3 2 5 1 3
> hs
AA AB AF DD CD BA
0.0804280653 0.0130094958 0.0047314403 0.0125549578 0.2350165253 0.0323901300
CB CC BC ED AE CA
0.7145888159 1.0000000000 0.2321828382 0.0485835689 0.0794970773 0.0449495643
EB BF BB AC DC BD
0.1870058122 0.1272314500 0.1150808240 0.2093517929 0.0074555153 0.0130595971
DB CF DF BE EA CE
0.0173817606 0.0071028158 0.0250578677 0.2195271464 0.0569340883 0.0631908504
EE EF FF FD GB GC
0.0127382402 0.0011190982 0.0044854582 0.0004563724 0.0003594258 0.0002507046
GD AD KA KF LC DA
0.0063744634 0.0096431701 0.0627646253 0.0004418478 0.0568163870 0.0000000000
EC FA FB DE FC FE
0.0000000000 0.0000000000 0.0000000000 0.0000000000 0.0000000000 0.0000000000
GA GE KB KC KD KE
0.0000000000 0.0000000000 0.0000000000 0.0000000000 0.0000000000 0.0000000000
LB LA LD LE
0.0000000000 0.0000000000 0.0000000000 0.0000000000

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> as
      AA      AB      AF      DD      CD      BA
4.412541e-03 3.968841e-03 8.375004e-02 1.409698e-01 7.894656e-02 2.146204e-02
      CB      CC      BC      ED      AE      CA
3.660444e-03 1.121941e-01 6.326094e-03 4.799095e-02 1.524499e-02 1.000000e+00
      EB      BF      BB      AC      DC      BD
0.000000e+00 6.072030e-03 8.323026e-02 6.262068e-02 0.000000e+00 2.486679e-04
      DB      CF      DF      BE      EA      CE
0.000000e+00 8.146610e-03 1.925650e-01 1.095155e-01 7.776767e-03 2.437300e-02
      EE      EF      FF      FD      GB      GC
2.419075e-02 6.390996e-05 0.000000e+00 2.071608e-03 0.000000e+00 0.000000e+00
      GD      AD      KA      KF      LC      DA
0.000000e+00 1.806051e-02 0.000000e+00 2.156359e-03 0.000000e+00 3.583779e-03
      EC      FA      FB      DE      FC      FE
2.468362e-02 8.227873e-03 3.500976e-02 4.973358e-04 2.815768e-03 3.064436e-03
      GA      GE      KB      KC      KD      KE
1.391587e-02 3.500976e-02 3.500976e-02 2.486679e-04 8.227873e-03 2.501759e-02
      LB      LA      LD      LE
8.227873e-03 1.133971e-03 1.656466e-04 1.133971e-03
>
> |
```